

Protein Epitope Scan – HLA-A*02:01

Model: Deep FFN (MHCflurry-trained, 91.9% accuracy, 89.6% CV)

Allele: HLA-A*02:01

Method: Sliding 9-mer window, BLOSUM62 encoding

Thresholds: SB = $P(\text{SB}) > 50\%$, WB = $P(\text{WB}) > 50\%$, NB otherwise

Date: June 15, 2026

Known Epitope Validation

Protein	Epitope	Pred	P(SB)	Rank	Status	Note
p53	LLGRNSFEV	SB	99.3%	Top 1.0%	[OK] Confirmed	Classic p53 265-273 epitope
Spike	YLQPRTFLL	SB	99.8%	–	[OK] Confirmed	Spike 269-277 (NTD, not RBD)
MART-1	ILTVILGVL	WB	3.3% (95.7% WB)	Top 4.5%	[OK] Confirmed	MART-1 32-40
Spike	RLFRKSNLK	NB	0%	Top 18%	[–] Missed	MHC class II epitope
MART-1	ELAGIGILTV	NB	0%	–	[!!] Missed	Clinical anchor-modified variant

Top Epitope Candidates per Protein

MART-1 / Melan-A (118 amino acids, melanoma antigen)

110 9-mer windows scanned | 2 SB, 12 WB, 96 NB

Pos	Peptide	Pred	P(SB)	P(WB)	Binding Score	Anchors
56-64	ALMDKSLHV	SB	100.0%	0.0%	1.000	[OK] L-V
31-39	GILTVILGV	SB	99.7%	0.3%	0.999	[OK] I-V
35-43	VILGVLLLI	SB	93.1%	6.9%	0.965	[OK] I-I
34-42	TVILGVLLL	WB	5.5%	91.5%	0.513	[OK] V-L
32-40	ILTVILGVL	WB	3.3%	95.7%	0.512	[OK] L-L
27-35	AAGIGILTV	WB	0.8%	97.2%	0.494	[OK] A-V
88-96	LQEKNCPEV	WB	0.1%	96.6%	0.484	[OK] Q-V
29-37	GIGILTVIL	WB	0.6%	95.4%	0.483	[OK] I-L
40-48	LLIGCWYC	WB	0.1%	93.7%	0.469	–
57-65	LMDKSLHVG	WB	0.2%	89.0%	0.447	–

SARS-CoV-2 Spike RBD (318 amino acids, receptor binding domain)

310 9-mer windows scanned | 5 SB, 27 WB, 278 NB

Pos	Peptide	Pred	P(SB)	P(WB)	Binding Score	Anchors
87-95	KIADYNYKL	SB	99.9%	0.1%	0.999	[OK] I-L
182-190	VLSFELLHA	SB	90.7%	9.3%	0.953	[OK] L-A
186-194	ELLHAPATV	SB	84.9%	15.1%	0.925	[OK] L-V
282-290	YQDVNCTEV	WB	35.1%	64.8%	0.676	[OK] Q-V
44-52	FSTFKCYGV	WB	17.0%	82.4%	0.582	–
172-180	GVGYQPYRV	WB	12.5%	86.3%	0.556	[OK] V-V
304-312	RVYSTGSNV	WB	4.5%	95.3%	0.521	[OK] V-V
114-122	KVGGNYYNL	WB	2.4%	96.9%	0.508	[OK] V-L
94-102	KLPDDFTGC	WB	1.7%	98.2%	0.508	–
175-183	YQPYRVVVL	WB	1.9%	97.3%	0.505	[OK] Q-L

p53 Tumor Suppressor (323 amino acids)

315 9-mer windows scanned | 8 SB, 25 WB, 282 NB

Pos	Peptide	Pred	P(SB)	P(WB)	Binding Score	Anchors
65-73	RMPEAAPPV	SB	100.0%	0.0%	1.000	[OK] M-V
164-172	KQSQHMTEV	SB	99.4%	0.6%	0.997	[OK] Q-V
264-272	LLGRNSFEV	SB	99.3%	0.7%	0.996	[OK] L-V
187-195	GLAPPQHLL	SB	97.9%	2.1%	0.990	[OK] L-I
129-137	ALNKMFCQL	SB	96.2%	3.8%	0.981	[OK] L-L
103-111	YQGSYGFRL	SB	94.2%	5.8%	0.971	[OK] Q-L

Pos	Peptide	Pred	P(SB)	P(WB)	Binding Score	Anchors
31-39	VLSPLPSQA	SB	92.0%	8.0%	0.960	[OK] L-A
132-140	KMFCQLAKT	SB	80.2%	19.8%	0.901	[OK] M-T
139-147	KTCPVQLWV	WB	39.2%	60.6%	0.696	[OK] T-V
135-143	CQLAKTCPV	WB	38.5%	61.1%	0.691	[OK] Q-V

Scan Statistics

Metric	MART-1	Spike RBD	p53	Total
Protein length (aa)	118	318	323	759
9-mer windows	110	310	315	772
Strong Binders	2 (1.8%)	5 (1.6%)	7 (2.2%)	14 (1.8%)
Weak Binders	12 (10.9%)	27 (8.7%)	25 (7.9%)	64 (8.3%)
Non-Binders	96 (87.3%)	278 (89.7%)	283 (89.8%)	694 (89.9%)

Key Findings

Validation Success

- **3 of 4 MHC-I epitopes confirmed** – 75% sensitivity on known positives
- p53 LLGRNSFEV correctly identified as SB at Top 1.0% rank
- MART-1 ILTVILGVL correctly identified as WB at Top 4.5% rank
- RLFRKSNLK missed because it is an MHC-II epitope (the model is MHC-I specific)

Novel Candidate Epitopes

Candidate	Protein	Confidence	Rationale
ALMDKSLHV	MART-1 56-64	100% SB	L-V anchors, hydrophobic core
KIADYNYKL	Spike RBD 87-95	99.9% SB	I-L anchors, in RBD core

Candidate	Protein	Confidence	Rationale
RMPEAAPPV	p53 65-73	100% SB	M-V anchors, known mutation hotspot
KQSQHMTEV	p53 164-172	99.4% SB	Q-V anchors, DNA binding domain
GLAPPQHLI	p53 187-195	97.9% SB	L-I anchors, proline-rich region

Anchor Residue Analysis

The model strongly weights canonical anchor positions: - **p2**: L, M, I, V, Q, A high SB probability - **p9**: V, L, I, A, T, M high SB probability - **Non-canonical p2** (E, D, R, K, G, P) almost always NB - **Cysteine at p2** occasionally tolerated in specific contexts

Clinical Relevance

- p53 LLGRNSFEV (264-272) is a validated immunotherapy target in multiple cancers
- p53 RMPEAAPPV (65-73) lies in the transactivation domain – frequently mutated
- Spike YLQPRTFLL (269-277) is the immunodominant CD8+ epitope in COVID-19
- The novel candidates should be validated by *in vitro* binding assays before clinical use

Methods

1. **Model**: Deep Feed-Forward Neural Network (36018090453)
2. **Training**: MHCflurry 2.2.0-labeled peptides, 5-fold CV 89.6% 0.8%
3. **Encoding**: BLOSUM62 substitution matrix (9-mer 920 tensor)
4. **Scan**: Sliding window (stride=1) across full protein sequences
5. **Thresholding**: SB = $P(\text{SB}) > P(\text{WB})$ and $P(\text{SB}) > P(\text{NB})$

Output Files

File	Description
data/protein_epitope_scan_full_scan	Full scan results (772 peptides with predictions)
protein_epitope_scan.md	This report

File	Description
plots/comparison_*.png	Model comparison visualizations

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